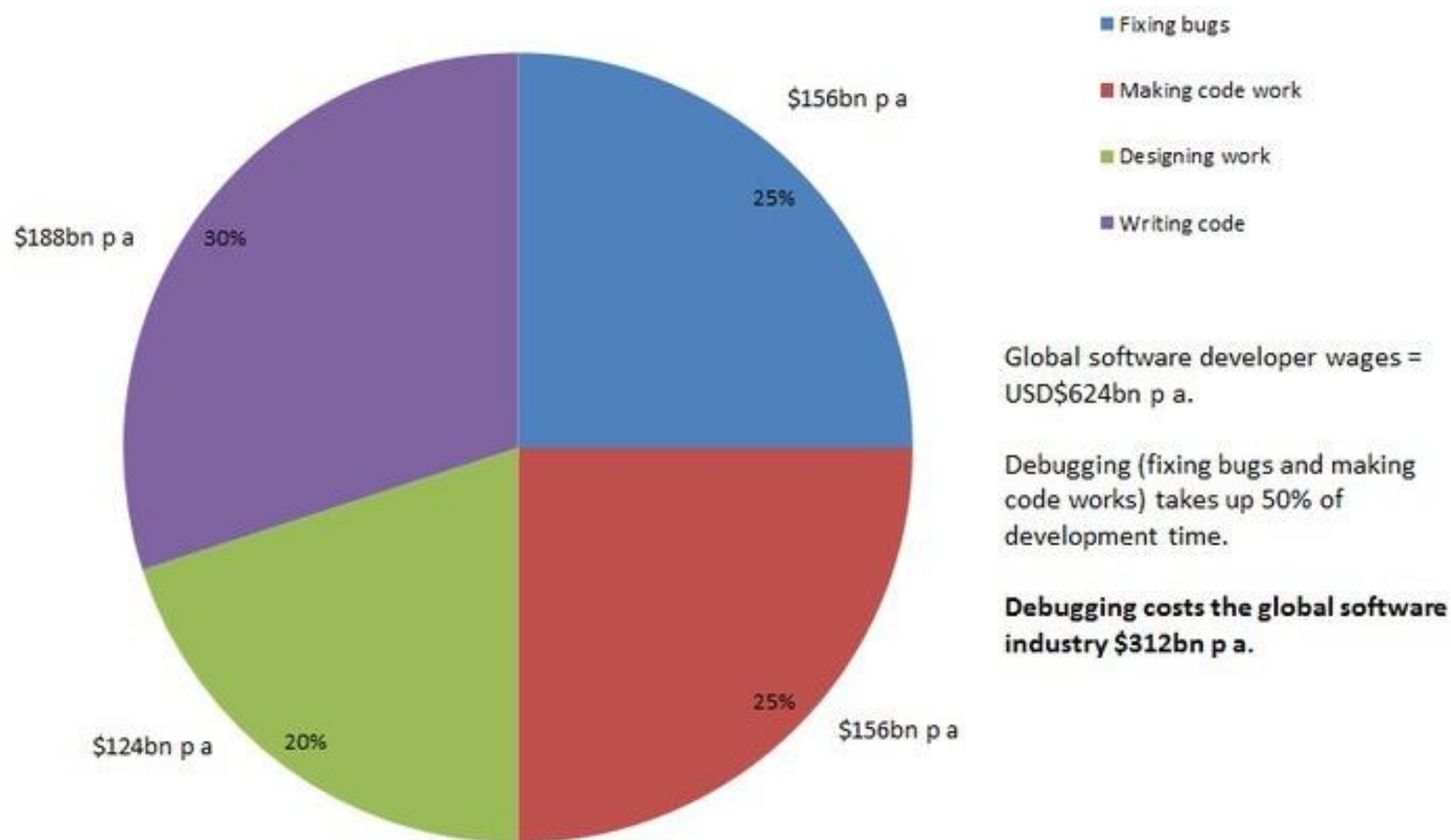


Отладка (debugging)

Павел Галанин
2025



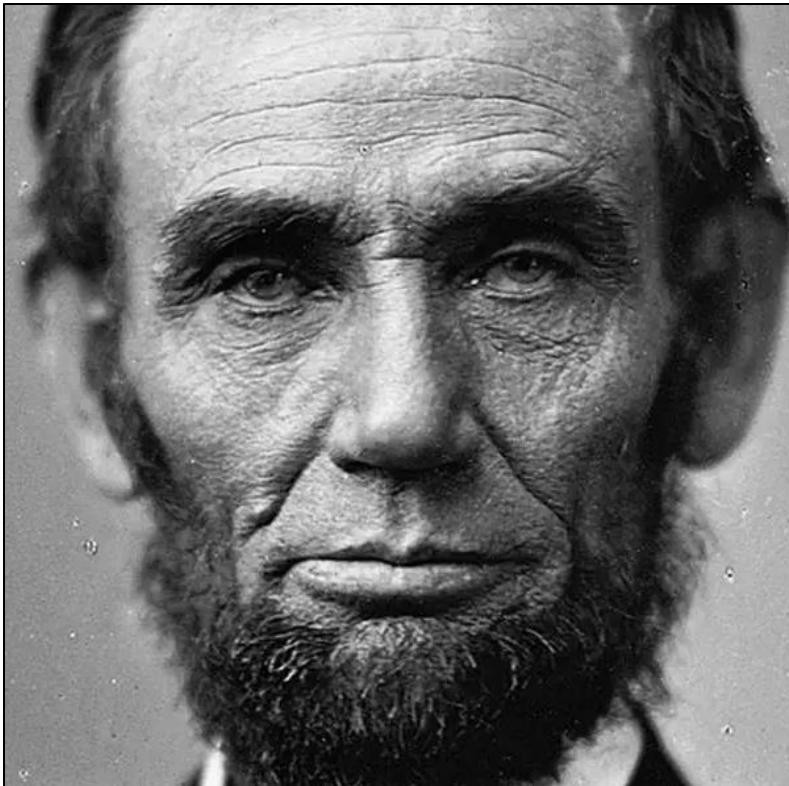
Impact of debugging on time spent developing code and its cost in terms of wages per annum





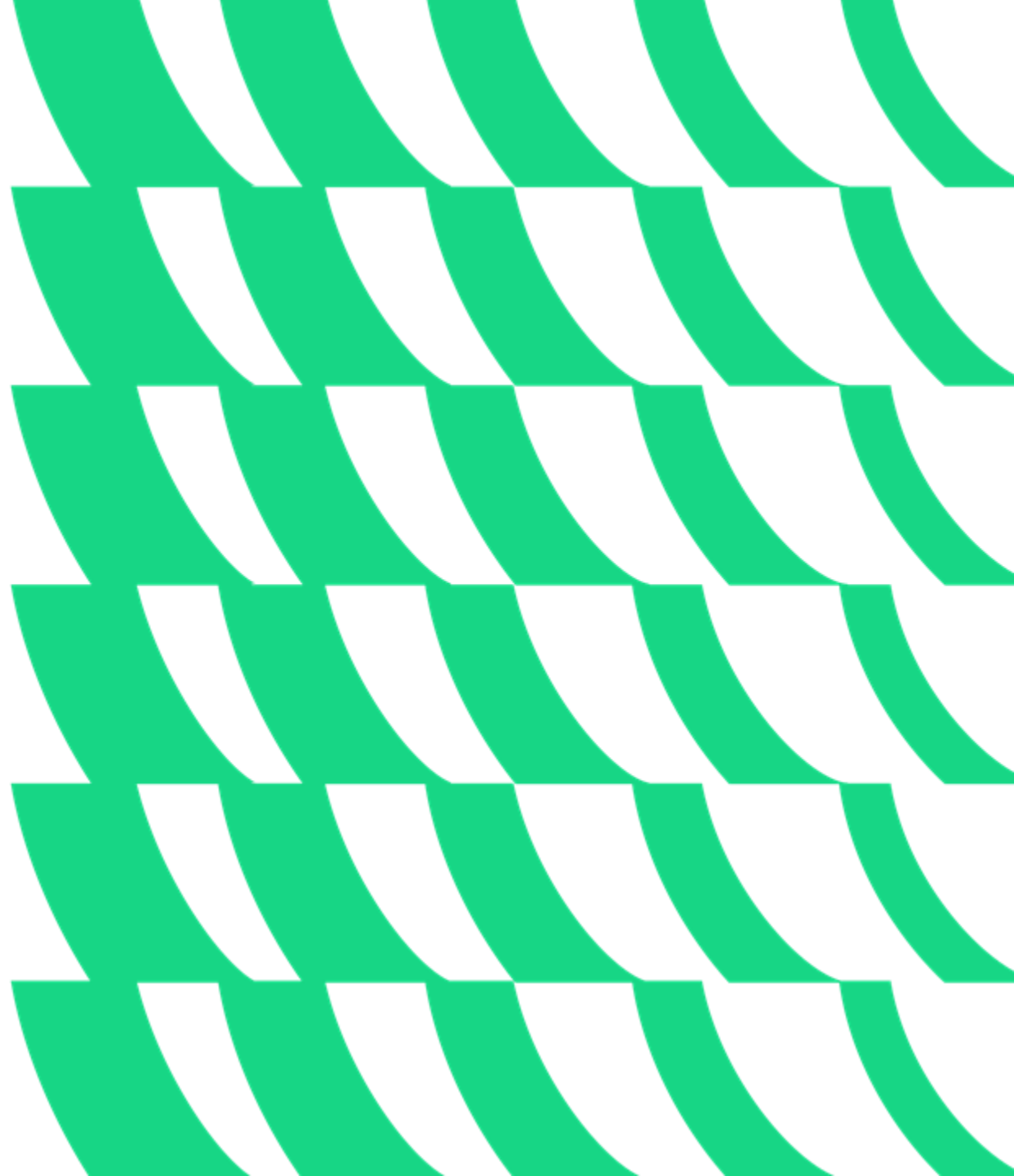
Give me six hours to chop down a tree and
I will spend

the first four sharpening
the axe.



Abraham Lincoln

Android Debug Bridge

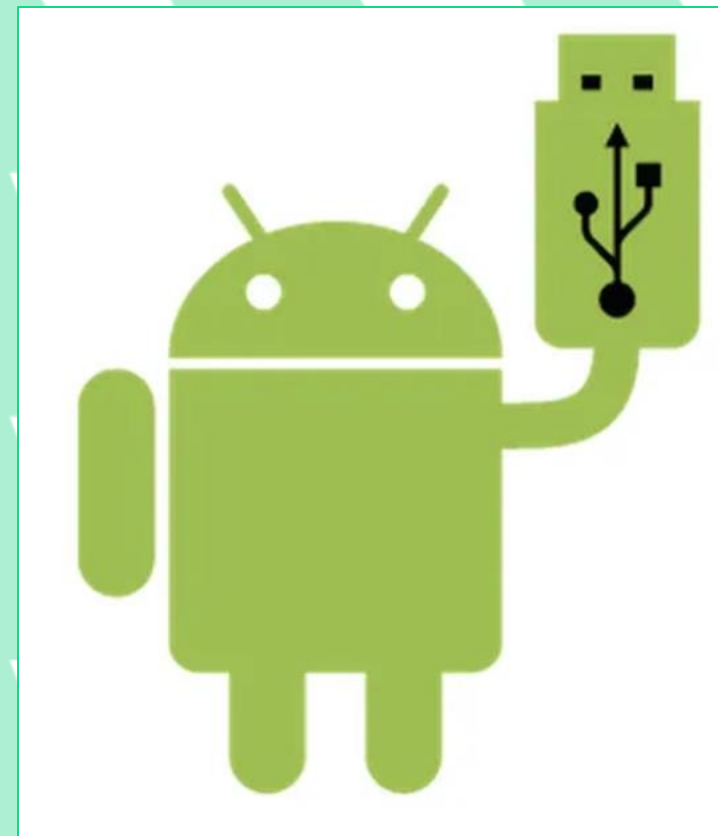


Android Debug Bridge (adb)

Что можно делать с помощью adb:

- устанавливать и удалять приложение
- читать логи приложения (logcat)
- передавать и скачивать файлы
- проксировать сетевой трафик
- получить доступ к shell на устройстве
- просмотр и изменения состояния устройства

...

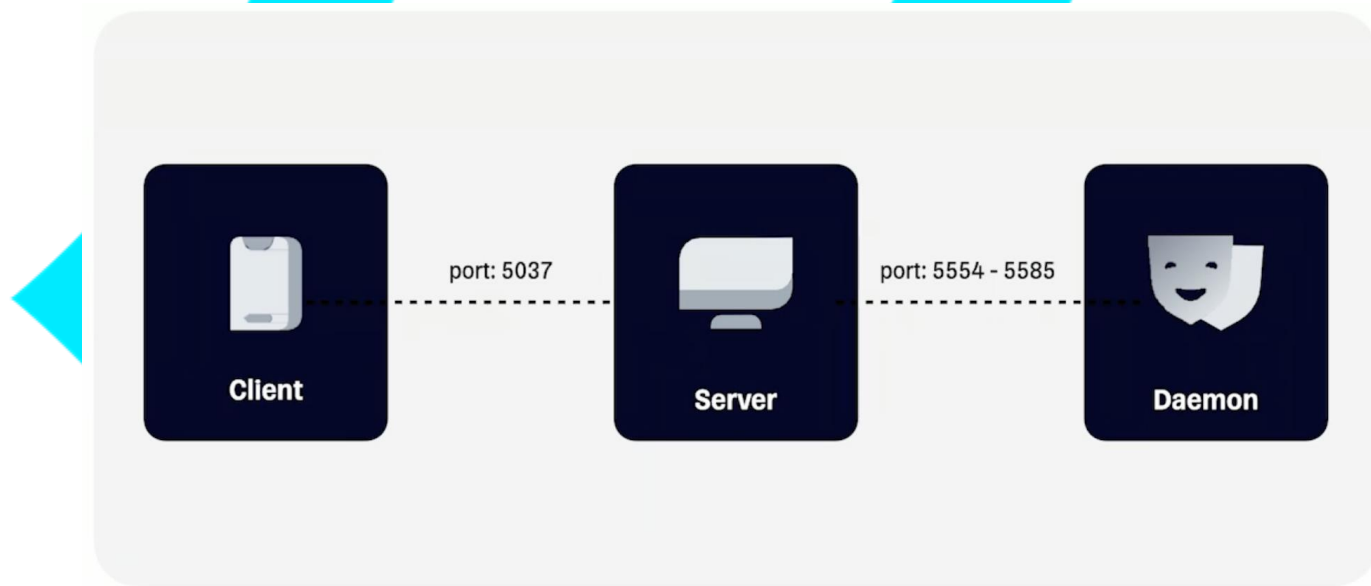


Как работает adb

Клиент - отправляет команды
Утилита на машине разработчика

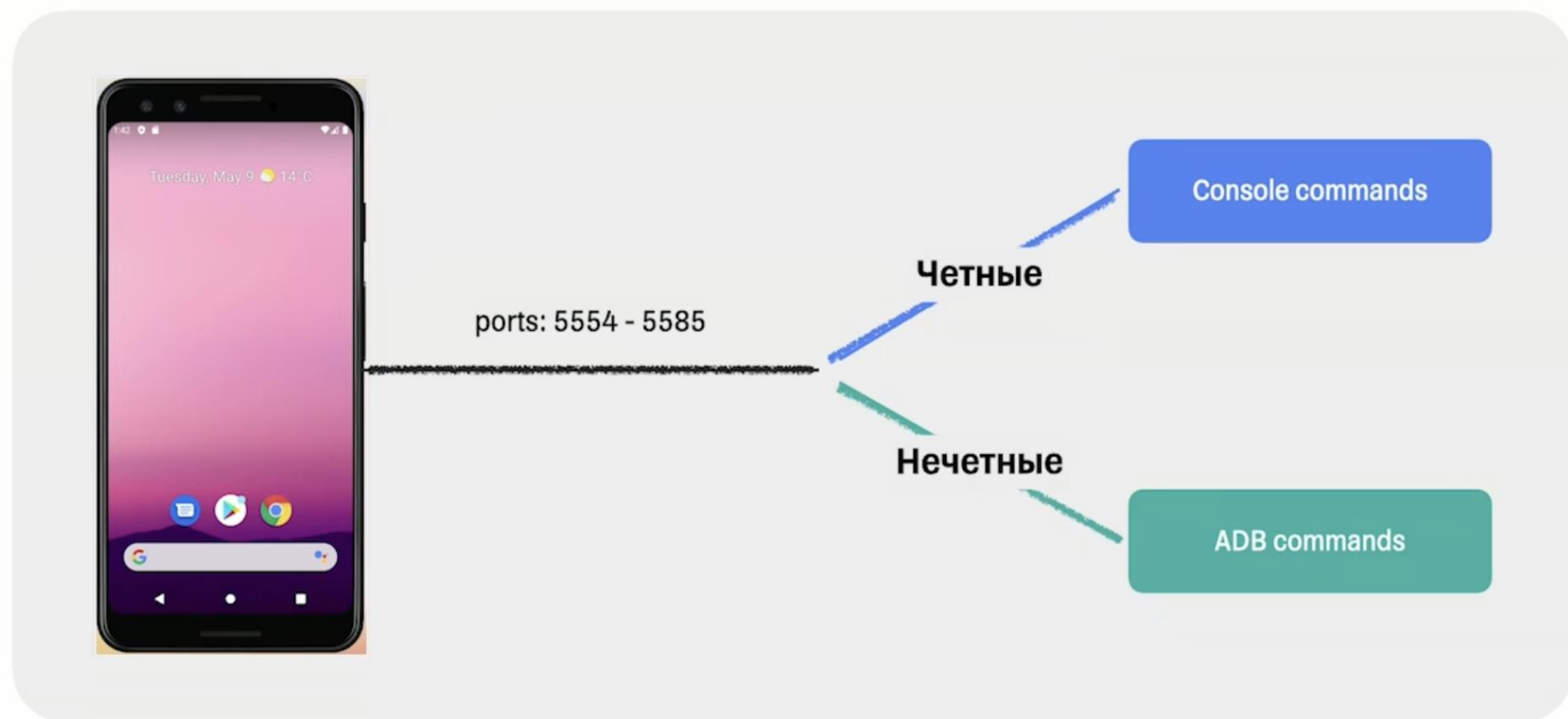
Демон `adbd` - выполняет команды на устройстве
Фоновый процесс на устройстве

Сервер - обеспечивает связь между клиентом и демоном
*Фоновый процесс на машине разработчика
Занимает TCP-порт 5037*



Как работает adb

Для каждого подключённого устройства подключается два порта:



Команды adb

список подключённых устройств

```
adb devices -l
```

установка/удаление приложения

```
adb install file_path
```

```
adb uninstall com.vk.debugging
```

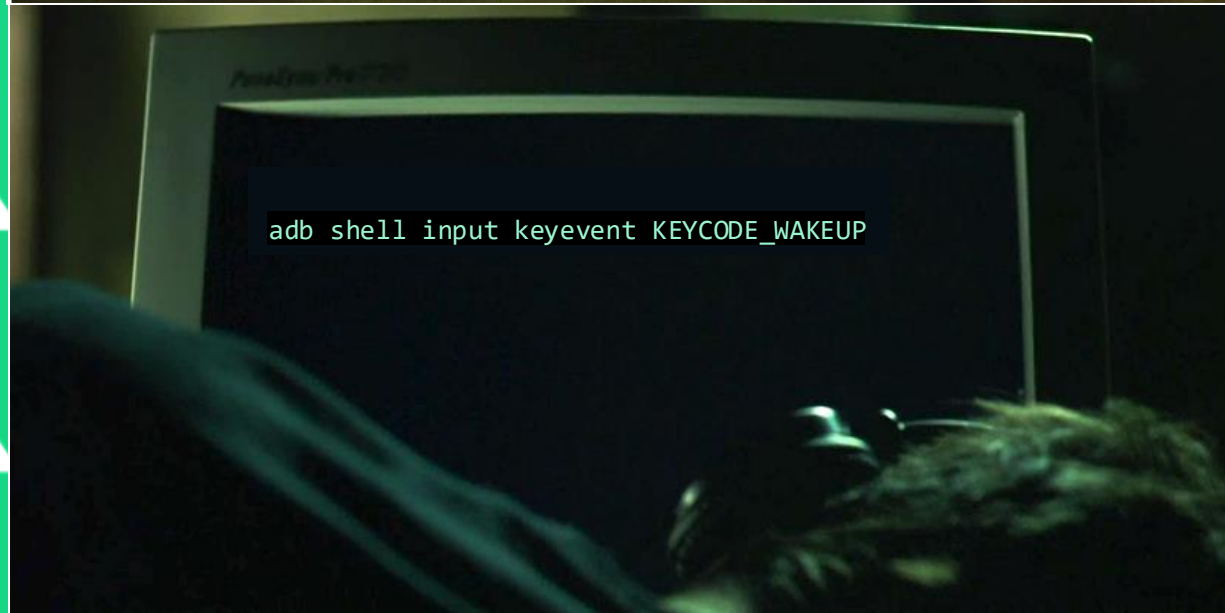
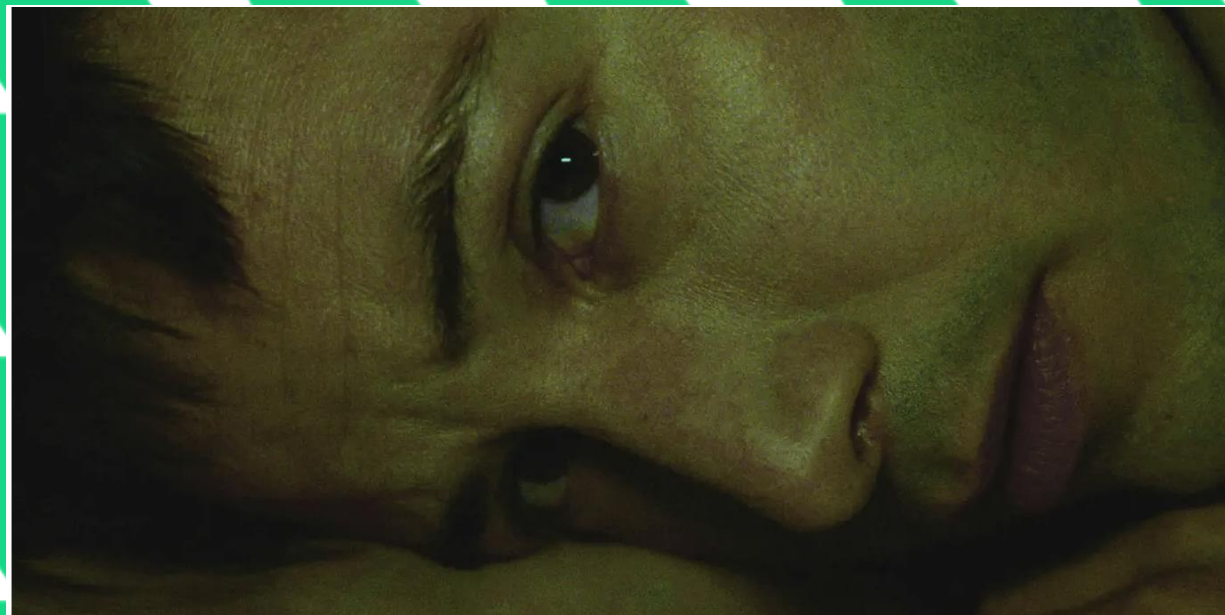
загрузка/скачивание файлов

```
adb pull remote local
```

```
adb push local remote
```

остановить сервер

```
adb kill-server
```



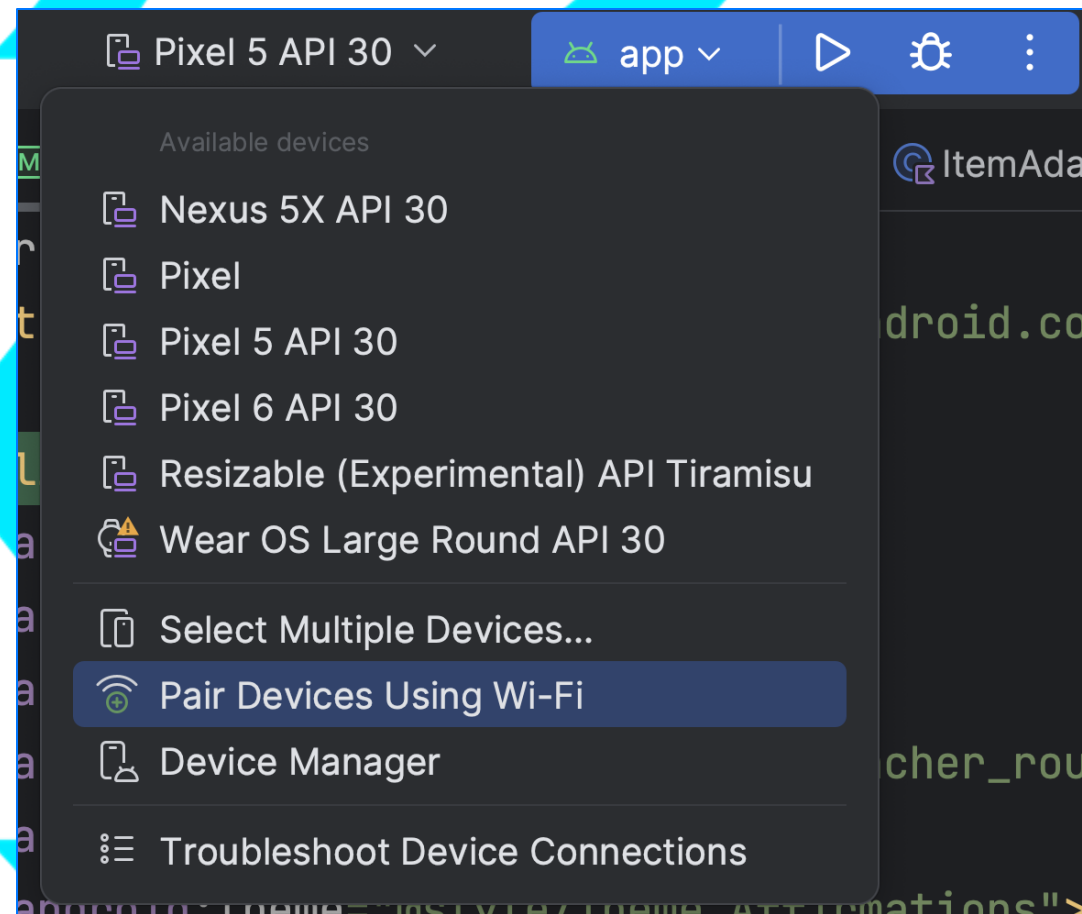
Подключение к устройству

USB

1. Включить в настройках разработчика USB debugging
2. Подключить устройство через кабель

Wi-Fi (*>= Android 11 - API level 30*)

1. Подключить устройство к той же сети, к какой подключена машина разработчика
2. В Android Studio: Pair Devices Using Wi-Fi.
3. На устройстве в настройках разработчика включить Wireless debugging
4. Соединить устройство через QR- или числовой код



adb shell

Перейти в интерактивный режим

```
adb shell
```

Посмотреть доступные shell-команды

```
adb shell ls /system/bin
```

Можно выполнить команду сразу

```
adb shell am start -a android.intent.action.VIEW
```

Ещё команды

```
adb shell am start -n com.vk.debugging/.MainActivity
```

```
adb shell am force-stop com.vk.debugging
```

```
adb shell pm clear com.vk.debugging
```

```
adb shell pm grant/revoke com.vk.debugging permission
```

PM

AM

SCREENCAP

SCREENRECORD

GETPROP/
SETPROP

START/STOP

INPUT

SERVICE

SVC

DUMPSYS

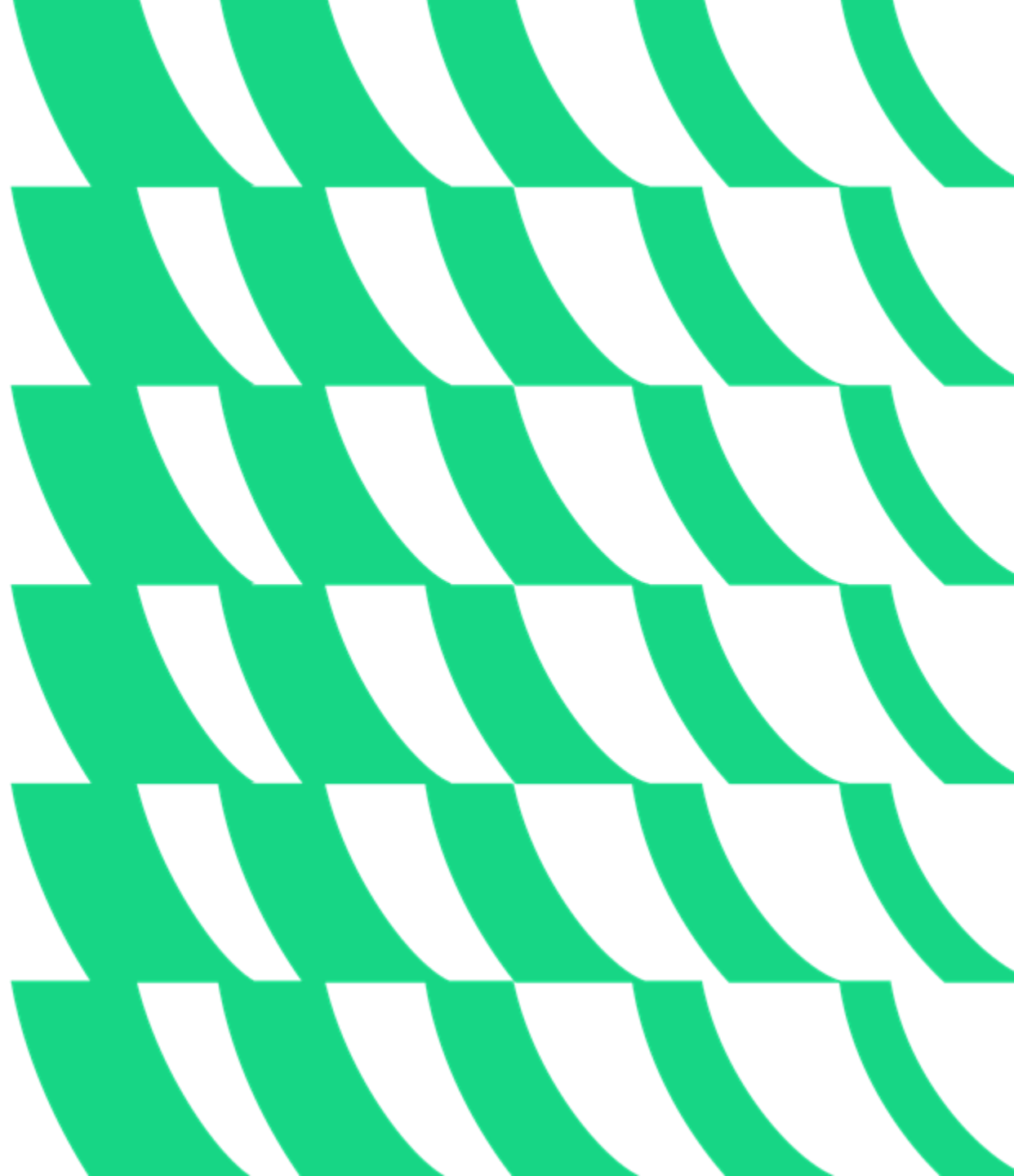
Logcat

- фильтрация по тегам, уровням, типам (креш, стектрейс, ...)
- логические выражения
- регулярные выражения
- ...

```
Logcat: Logcat x +
Pixel XL API 31 (emulator-5554) Android 12, API 31 package:mime

2022-12-29 01:00:55.790 1639-1855 EGL_emulation com.google.samples.apps.sunflower D app_time_stats: avg=27989792.00ms min=8.37ms max=55979576.00ms count=2
2022-12-29 01:01:04.770 1639-1675 .apps.sunflowe com.google.samples.apps.sunflower W Reducing the number of considered missed Gc histogram windows from 5600 to 100
2022-12-29 02:02:23.199 546-603 VerityUtils system_server E Failed to measure fs-verity, errno 1: /data/app/~~arQa19IZQnQgkPbxc1bWag==/com.google.samples.apps.sunflower-gPx95SCtI
2022-12-29 02:02:23.400 22470-22470 GraphicsEnvironment com.google.samples.apps.sunflower V ANGLE Developer option for 'com.google.samples.apps.sunflower' set to: 'default'
2022-12-29 02:02:23.400 22470-22470 com.google.samples.apps.sunflower V Neither updatable production driver nor prerelease driver is supported.
2022-12-29 02:02:23.402 22470-22470 NetworkSecurityConfig com.google.samples.apps.sunflower D No Network Security Config specified, using platform default
2022-12-29 02:02:23.402 22470-22470 com.google.samples.apps.sunflower D No Network Security Config specified, using platform default
2022-12-29 02:02:23.621 22470-22584 libEGL com.google.samples.apps.sunflower D loaded /vendor/lib64/egl/libEGL_emulation.so
2022-12-29 02:02:23.664 22470-22584 com.google.samples.apps.sunflower D loaded /vendor/lib64/egl/libGLESv1_CM_emulation.so
2022-12-29 02:02:23.673 22470-22584 com.google.samples.apps.sunflower D loaded /vendor/lib64/egl/libGLESv2_emulation.so
2022-12-29 02:02:23.743 546-603 VerityUtils system_server E Failed to measure fs-verity, errno 1: /data/app/~~arQa19IZQnQgkPbxc1bWag==/com.google.samples.apps.sunflower-gPx95SCtI
2022-12-29 02:02:24.327 22470-22470 .apps.sunflowe com.google.samples.apps.sunflower W Accessing hidden method Landroid/view/View;->computeFitSystemWindows(Landroid/graphics/Rect;Landroid/graphics/Rect;)Z
2022-12-29 02:02:24.328 22470-22470 com.google.samples.apps.sunflower W Accessing hidden method Landroid/view/ViewGroup;->makeOptionalFitsSystemWindows()V (unsupported, reflection, allowed)
2022-12-29 02:02:25.690 22470-22470 Compatibil...geReporter com.google.samples.apps.sunflower D Compat change id reported: 171228096; UID 10148; state: ENABLED
2022-12-29 02:02:26.155 22470-22470 Choreographer com.google.samples.apps.sunflower I Skipped 154 frames! The application may be doing too much work on its main thread.
2022-12-29 02:02:26.579 22470-22582 HostConnection com.google.samples.apps.sunflower D createUnique: call
2022-12-29 02:02:26.584 22470-22582 com.google.samples.apps.sunflower D HostConnection::get() New Host Connection established 0x7df95262cb90, tid 22582
2022-12-29 02:02:26.699 22470-22582 com.google.samples.apps.sunflower D HostComposition ext ANDROID_EMU_CHECKSUM_HELPER_v1 ANDROID_EMU_native_sync_v2 ANDROID_EMU_native_sync_v3 ANDROID_EMU_
2022-12-29 02:02:26.709 22470-22582 OpenGLRenderer com.google.samples.apps.sunflower W Failed to choose config with EGL_SWAP_BEHAVIOR_PRESERVED, retrying without...
2022-12-29 02:02:26.711 22470-22582 com.google.samples.apps.sunflower W Failed to initialize 101010-2 format, error = EGL_SUCCESS
2022-12-29 02:02:26.713 22470-22582 EGL_emulation com.google.samples.apps.sunflower D eglCreateContext: 0x7df95262cf50: maj 3 min 0 rcv 3
2022-12-29 02:02:26.715 22470-22582 com.google.samples.apps.sunflower D eglMakeCurrent: 0x7df95262cf50: ver 3 0 (tinfo 0x7dfb6e942080) (first time)
2022-12-29 02:02:26.803 22470-22582 Gralloc4 com.google.samples.apps.sunflower I mapper 4.x is not supported
```

Средства Android Studio



Logcat

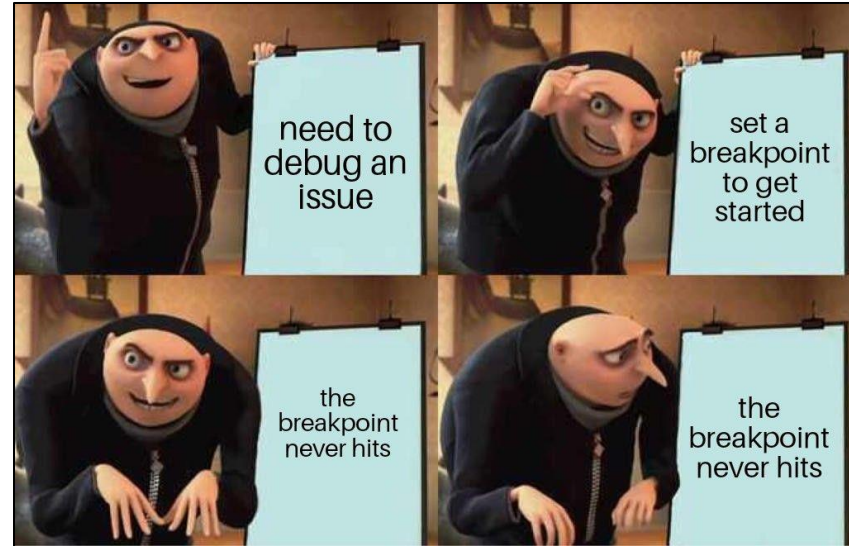
- фильтрация по тегам, уровням, типам (креш, стектрейс, ...)
- логические выражения
- регулярные выражения
- ...

```
Logcat: Logcat x +
Pixel XL API 31 (emulator-5554) Android 12, API 31 package:mime

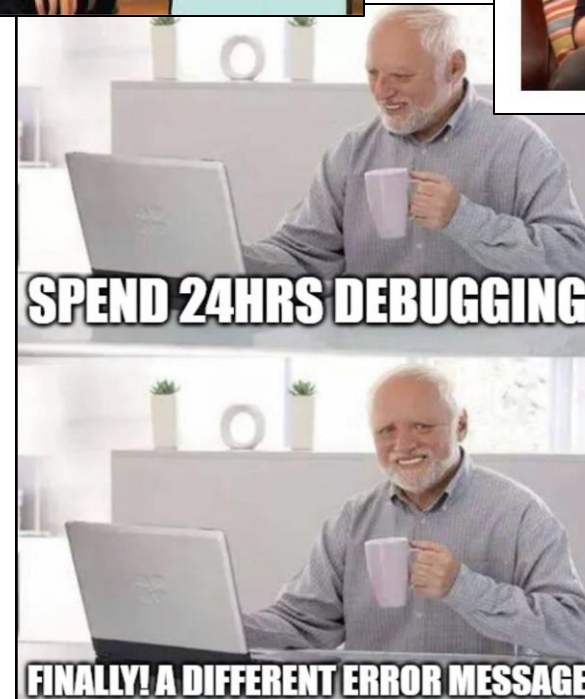
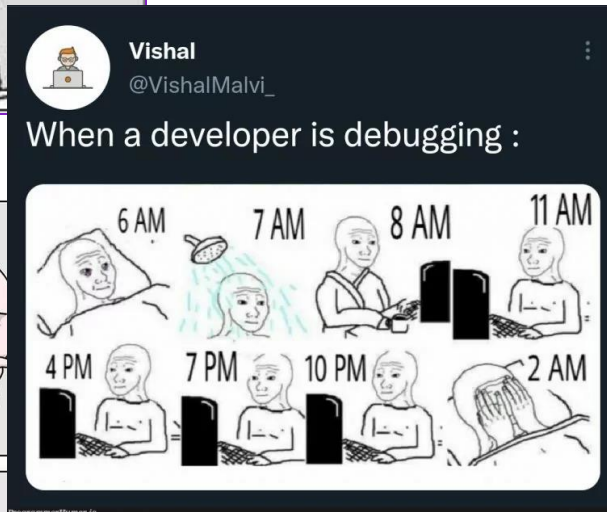
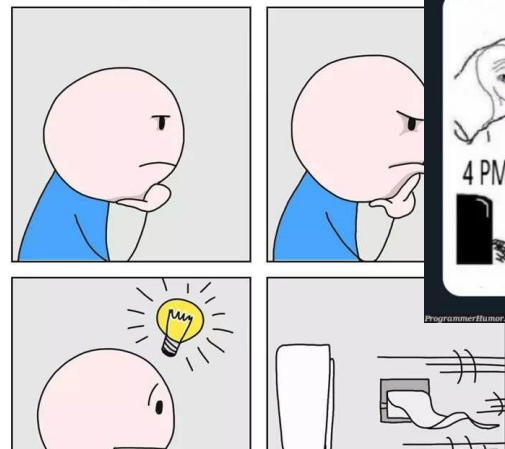
2022-12-29 01:00:55.790 1639-1855 EGL_emulation com.google.samples.apps.sunflower D app_time_stats: avg=27989792.00ms min=8.37ms max=55979576.00ms count=2
2022-12-29 01:01:04.770 1639-1675 .apps.sunflowe com.google.samples.apps.sunflower W Reducing the number of considered missed Gc histogram windows from 5600 to 100
2022-12-29 02:02:23.199 546-603 VerityUtils system_server E Failed to measure fs-verity, errno 1: /data/app/~~arQa19IZQnQgkPbxc1bWag==/com.google.samples.apps.sunflower-gPx95SCtI
2022-12-29 02:02:23.400 22470-22470 GraphicsEnvironment com.google.samples.apps.sunflower V ANGLE Developer option for 'com.google.samples.apps.sunflower' set to: 'default'
2022-12-29 02:02:23.400 22470-22470 com.google.samples.apps.sunflower V Neither updatable production driver nor prerelease driver is supported.
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2022-12-29 02:02:23.743 546-603 VerityUtils system_server E Failed to measure fs-verity, errno 1: /data/app/~~arQa19IZQnQgkPbxc1bWag==/com.google.samples.apps.sunflower-gPx95SCtI
2022-12-29 02:02:24.327 22470-22470 .apps.sunflowe com.google.samples.apps.sunflower W Accessing hidden method Landroid/view/View;->computeFitSystemWindows(Landroid/graphics/Rect;Landroid/graphics/Rect;)Z
2022-12-29 02:02:24.328 22470-22470 com.google.samples.apps.sunflower W Accessing hidden method Landroid/view/ViewGroup;->makeOptionalFitsSystemWindows()V (unsupported, reflection, allowed)
2022-12-29 02:02:25.690 22470-22470 Compatibil...geReporter com.google.samples.apps.sunflower D Compat change id reported: 171228096; UID 10148; state: ENABLED
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2022-12-29 02:02:26.699 22470-22582 com.google.samples.apps.sunflower D HostComposition ext ANDROID_EMU_CHECKSUM_HELPER_v1 ANDROID_EMU_native_sync_v2 ANDROID_EMU_native_sync_v3 ANDROID_EMU_
2022-12-29 02:02:26.709 22470-22582 OpenGLRenderer com.google.samples.apps.sunflower W Failed to choose config with EGL_SWAP_BEHAVIOR_PRESERVED, retrying without...
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2022-12-29 02:02:26.713 22470-22582 EGL_emulation com.google.samples.apps.sunflower D eglCreateContext: 0x7df95262cf50: maj 3 min 0 rcv 3
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2022-12-29 02:02:26.803 22470-22582 Gralloc4 com.google.samples.apps.sunflower I mapper 4.x is not supported
```

Directed by
ROBERT B. WEIDE

Отладка...



DEBUGGING



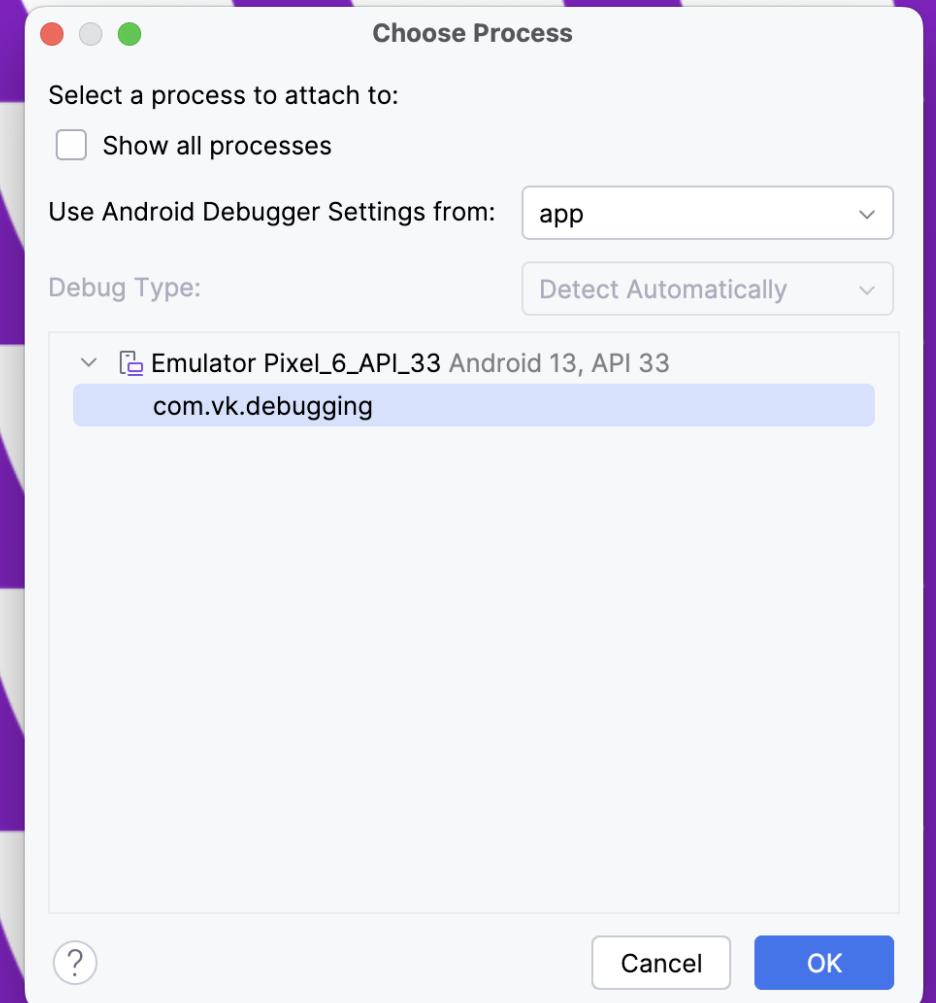
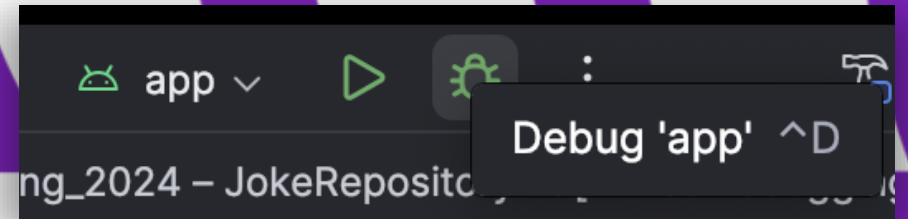
Breakpoint (точка останова)

```
19      }
20      jokeDto?.let {
21          localSource.saveJoke(it.mapToEntity())
22      }
23      localSource.loadJokes().last()
24  }
25  }
26  }
```

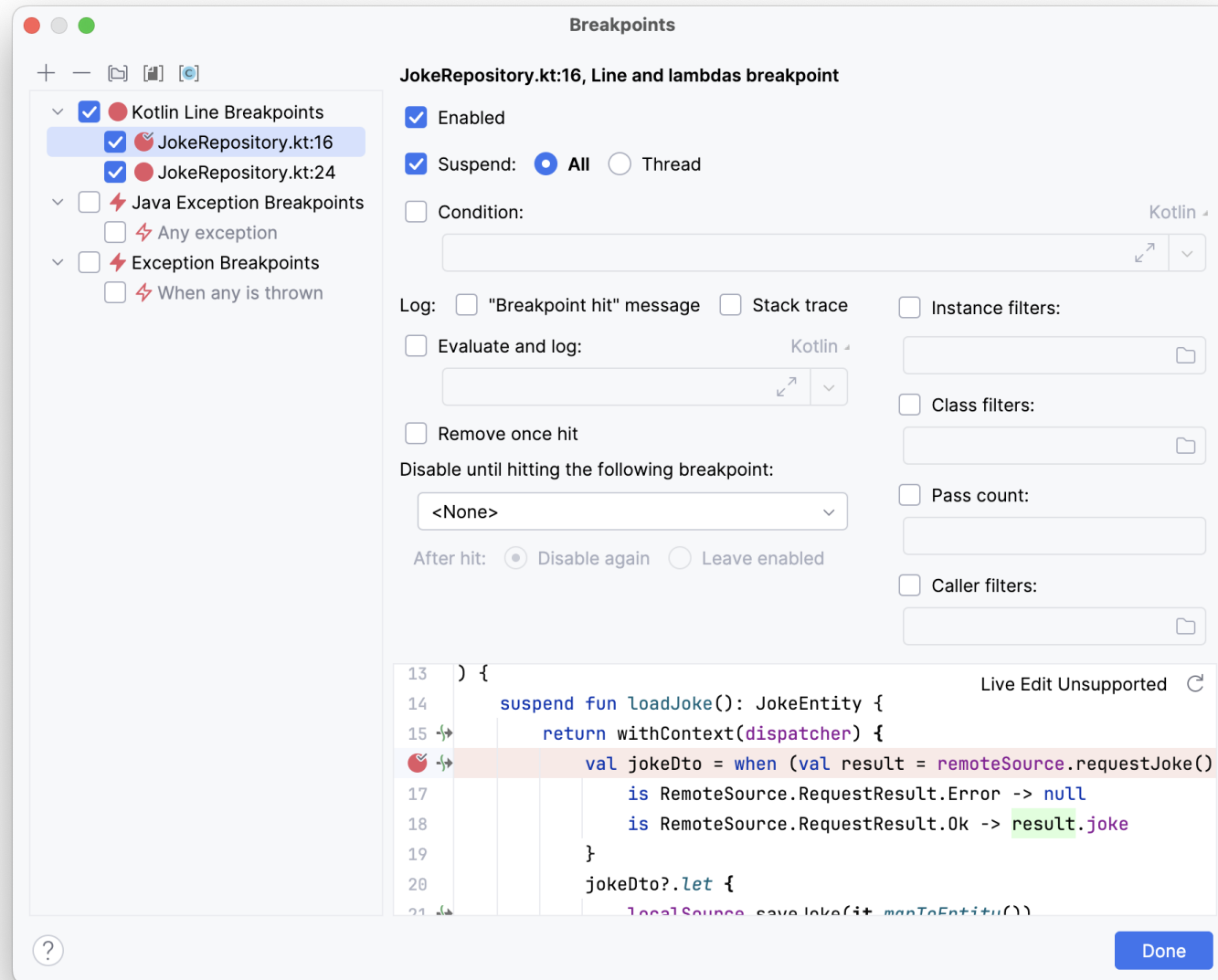
```
19      }
20      jokeDto?.let { jokeDto: JokeDto(joke=Chuck N
21          localSource.saveJoke(it.mapToEntity())
22      }
23      localSource.loadJokes().last()
24  }
25  }
```

Режим отладки

1. Запустить сразу в режиме отладки
2. Подсоединиться отладчиком к запущенному процессу



Настройки breakpoint



App Inspection: Database Inspector

App Inspection

Pixel 6 API 33 > com.vk.debugging

Database Inspector Network Inspector Background Task Inspector

Databases **joke**

Live updates

joke_db

- joke**
 - uid : INTEGER, NOT NULL
 - joke : TEXT, NOT NULL
 - url : TEXT, NOT NULL
- room_master_table

	uid	joke
1	1	Chuck Norris is so tough he makes onions cry.
2	2	Chuck Norris turned his Total Gym into a time machine.
3	3	When Chuck Norris throws a game of 'Washers' he uses a Maytag.
4	4	You will never see a Sasquatch because they're not about to risk running into Chuck Norris.
5	5	If Chuck Norris writes code with bugs, the bugs fix themselves.
6	6	When Chuck Norris speaks, everyone can understand him in their own tongue.
7	7	In a tagteam match, Chuck Norris was teamed with Hulk Hogan against King Kong Bundy.

App Inspection: Network Inspector

App Inspection

Pixel 6 API 33 > com.vk.debugging

Database Inspector

Network Inspector

Background Task Inspector

NETWORK
12 KB/s

00.000 05.000 10.000 15.000 20.000 25.000

Receiving: N/A Sending: N/A

Connection View

Thread View

Rules

Name	Size	Type	Status	Time	Timeline
random	268 B	json	200	188 ms	00.000 15.000

Overview

Response

Request

Call Stack

```
{ "categories": [], "created_at": "2020-01-05 13:42:20.8", "id": "Gyh3hE0nR0G_13:42:20.841843", "url": "https://api.chucknorris.io/said that a watch pot nevers boils, but in Chuck No out of fear and retaliation from Chuck Norris" }
```

Request

random

Method

GET

Status

200

Response type

application/json

Response size

268 B

code

app

src

main

java

com

vk

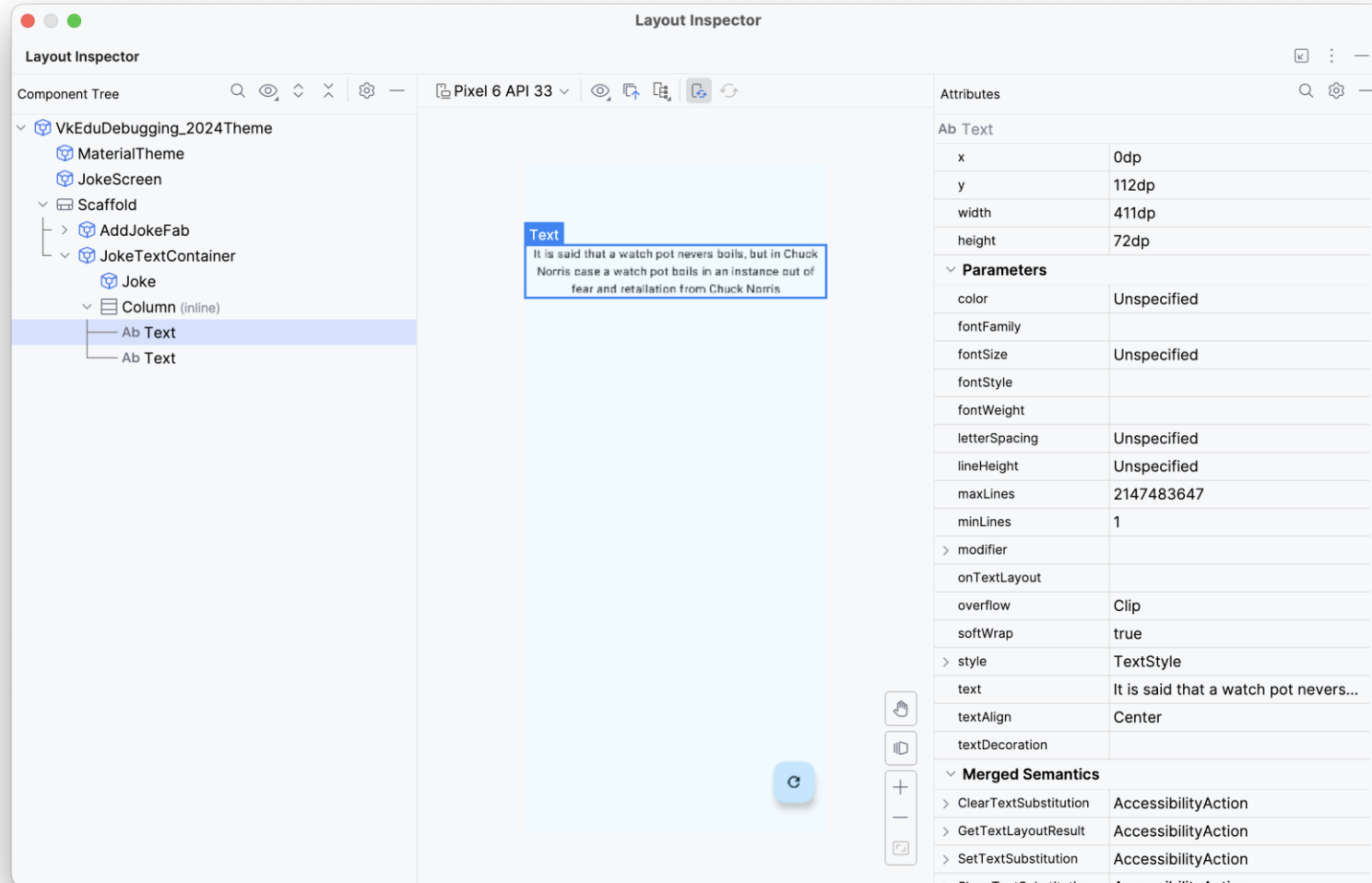
debugging

repo

JokeRepository

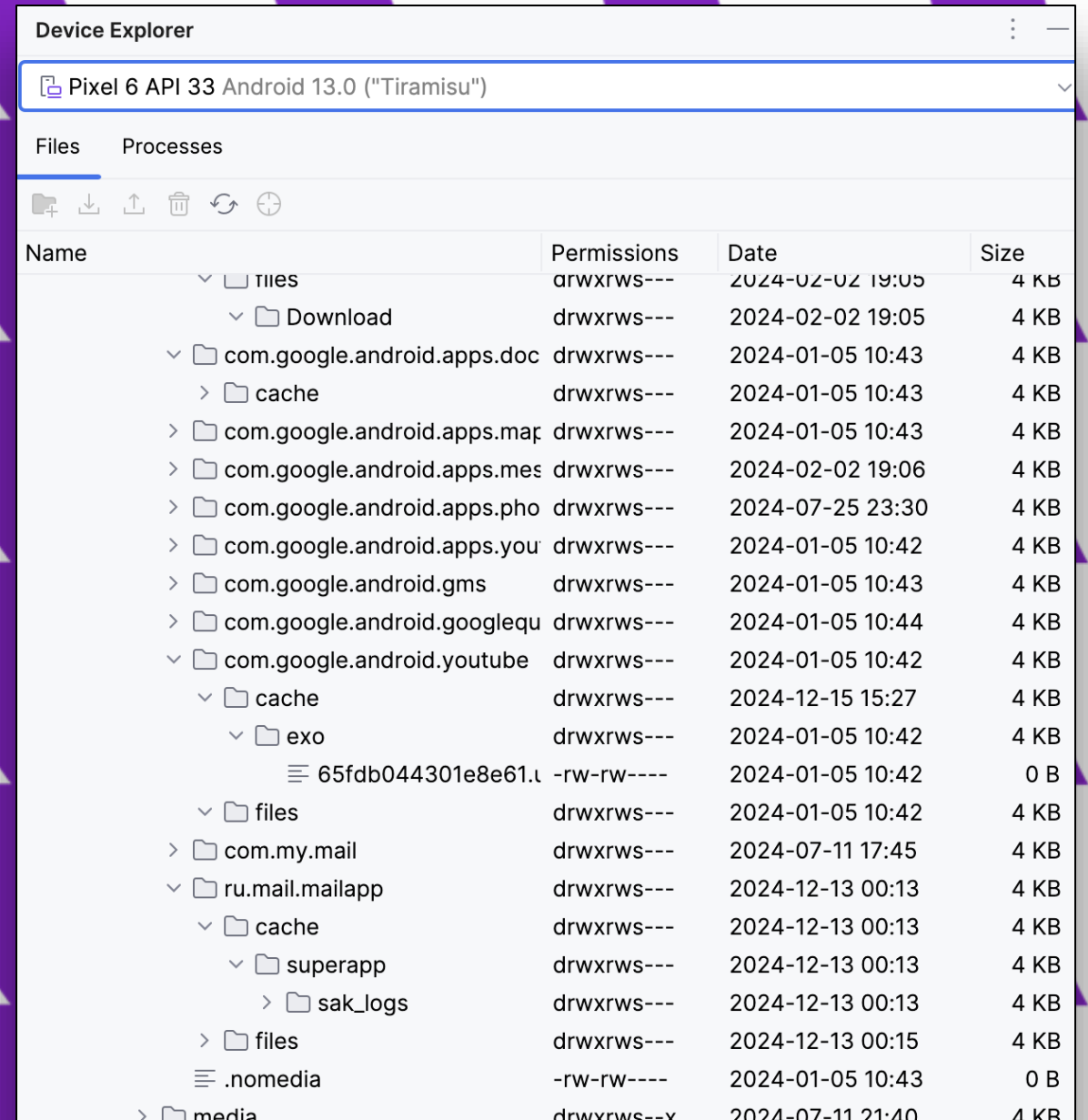
loadJoke: JokeEntity

Layout Inspector



File Inspector

Находится в разделе
Device Explorer

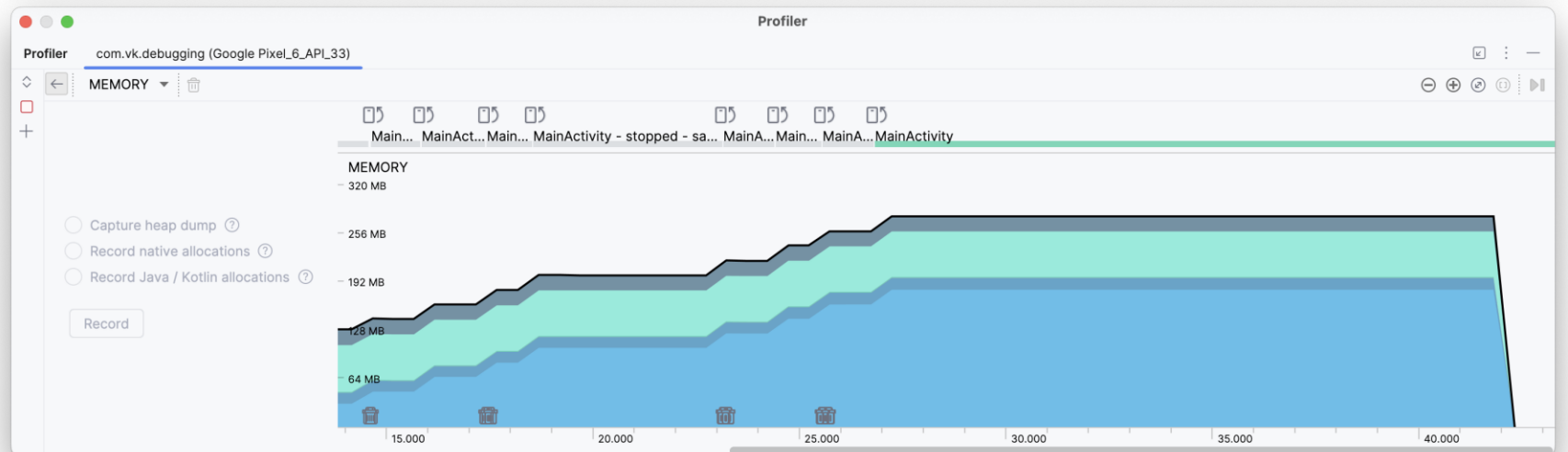
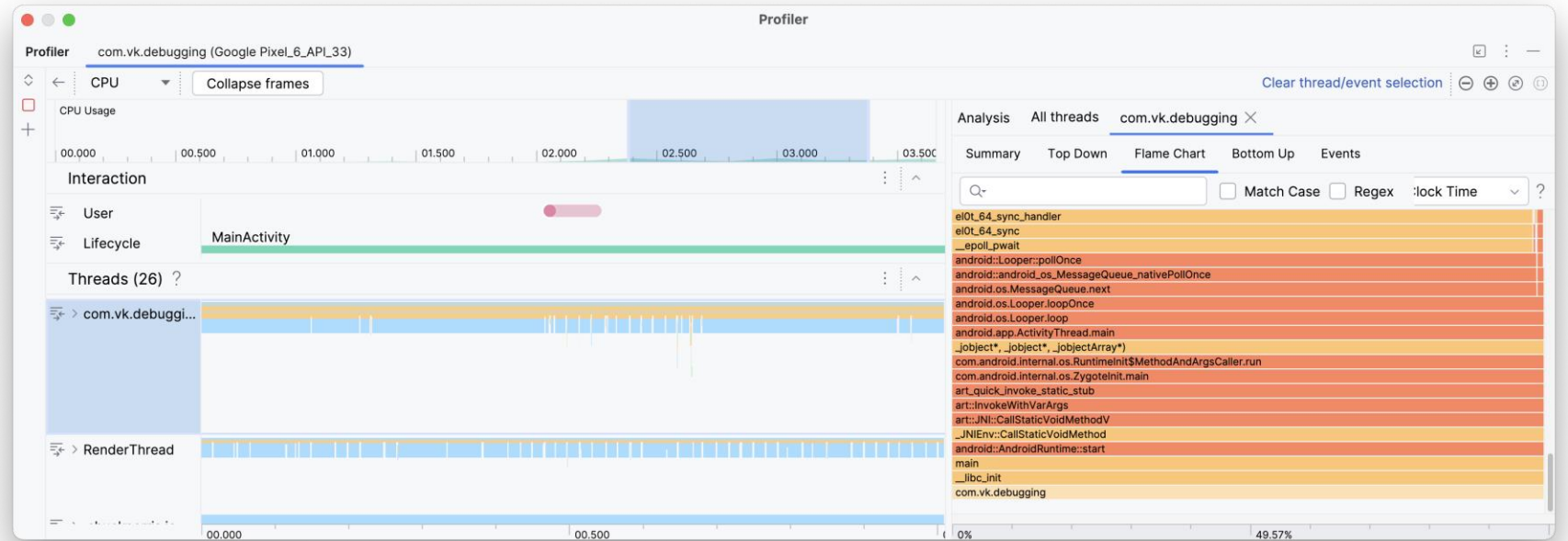


The screenshot shows the 'Device Explorer' window with the 'Files' tab selected. The title bar indicates the device is 'Pixel 6 API 33 Android 13.0 ("Tiramisu")'. Below the title bar, there are tabs for 'Files' and 'Processes', and a toolbar with icons for file operations. The main area displays a table of files and folders.

Name	Permissions	Date	Size
▼ [icon] Files	drwxrws---	2024-02-02 19:05	4 KB
▼ [icon] Download	drwxrws---	2024-02-02 19:05	4 KB
▼ [icon] com.google.android.apps.doc	drwxrws---	2024-01-05 10:43	4 KB
> [icon] cache	drwxrws---	2024-01-05 10:43	4 KB
> [icon] com.google.android.apps.mar	drwxrws---	2024-01-05 10:43	4 KB
> [icon] com.google.android.apps.mes	drwxrws---	2024-02-02 19:06	4 KB
> [icon] com.google.android.apps.pho	drwxrws---	2024-07-25 23:30	4 KB
> [icon] com.google.android.apps.you	drwxrws---	2024-01-05 10:42	4 KB
> [icon] com.google.android.gms	drwxrws---	2024-01-05 10:43	4 KB
> [icon] com.google.android.googlequ	drwxrws---	2024-01-05 10:44	4 KB
▼ [icon] com.google.android.youtube	drwxrws---	2024-01-05 10:42	4 KB
▼ [icon] cache	drwxrws---	2024-12-15 15:27	4 KB
▼ [icon] exo	drwxrws---	2024-01-05 10:42	4 KB
≡ 65fdb044301e8e61.l	-rw-rw----	2024-01-05 10:42	0 B
▼ [icon] files	drwxrws---	2024-01-05 10:42	4 KB
> [icon] com.my.mail	drwxrws---	2024-07-11 17:45	4 KB
▼ [icon] ru.mail.mailapp	drwxrws---	2024-12-13 00:13	4 KB
▼ [icon] cache	drwxrws---	2024-12-13 00:13	4 KB
▼ [icon] superapp	drwxrws---	2024-12-13 00:13	4 KB
> [icon] sak_logs	drwxrws---	2024-12-13 00:13	4 KB
> [icon] files	drwxrws---	2024-12-13 00:15	4 KB
≡ .nomedia	-rw-rw----	2024-01-05 10:43	0 B
> [icon] media	drwxrws---	2024-07-11 21:40	4 KB

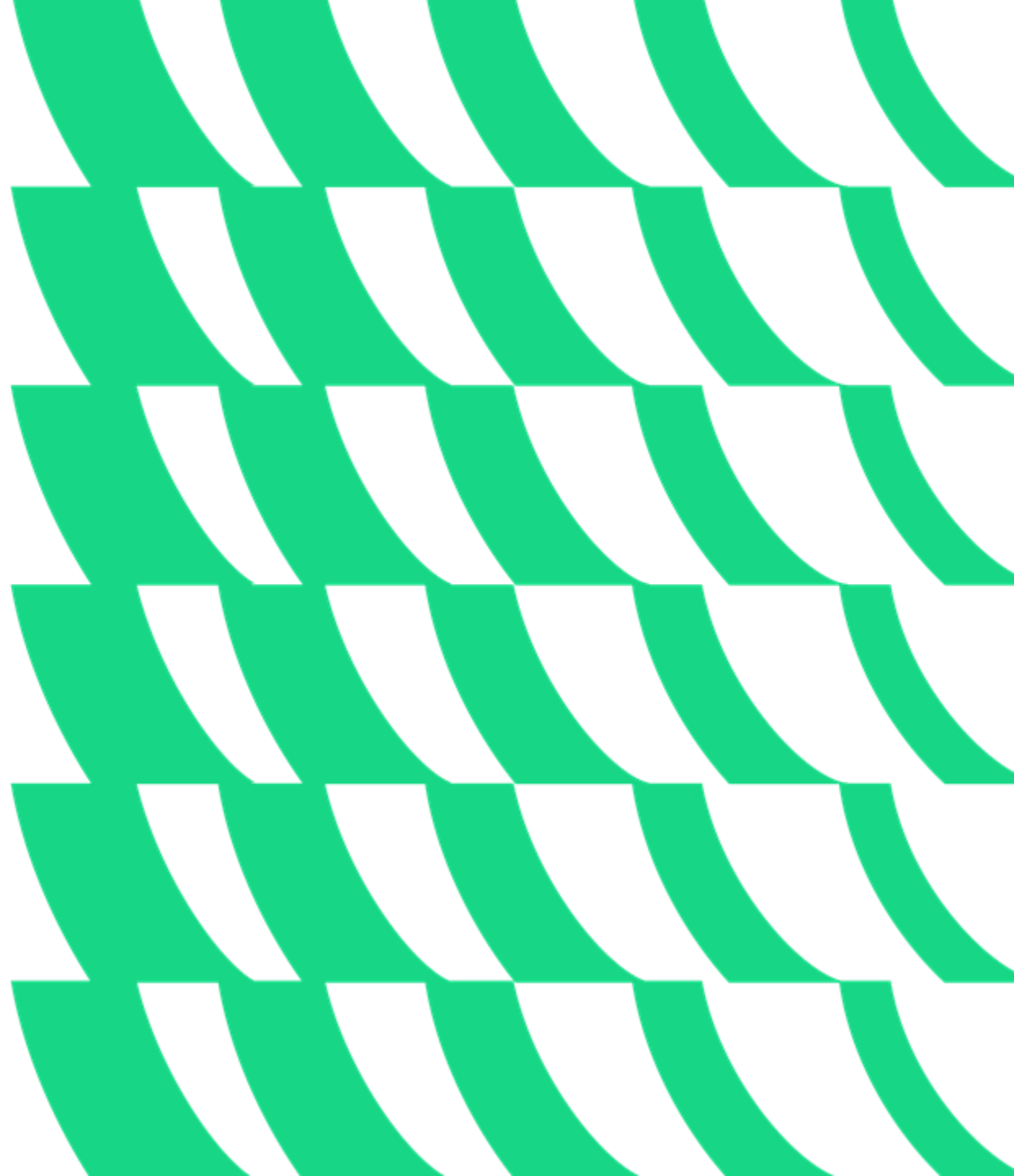
Profiler

- CPU
- Memory

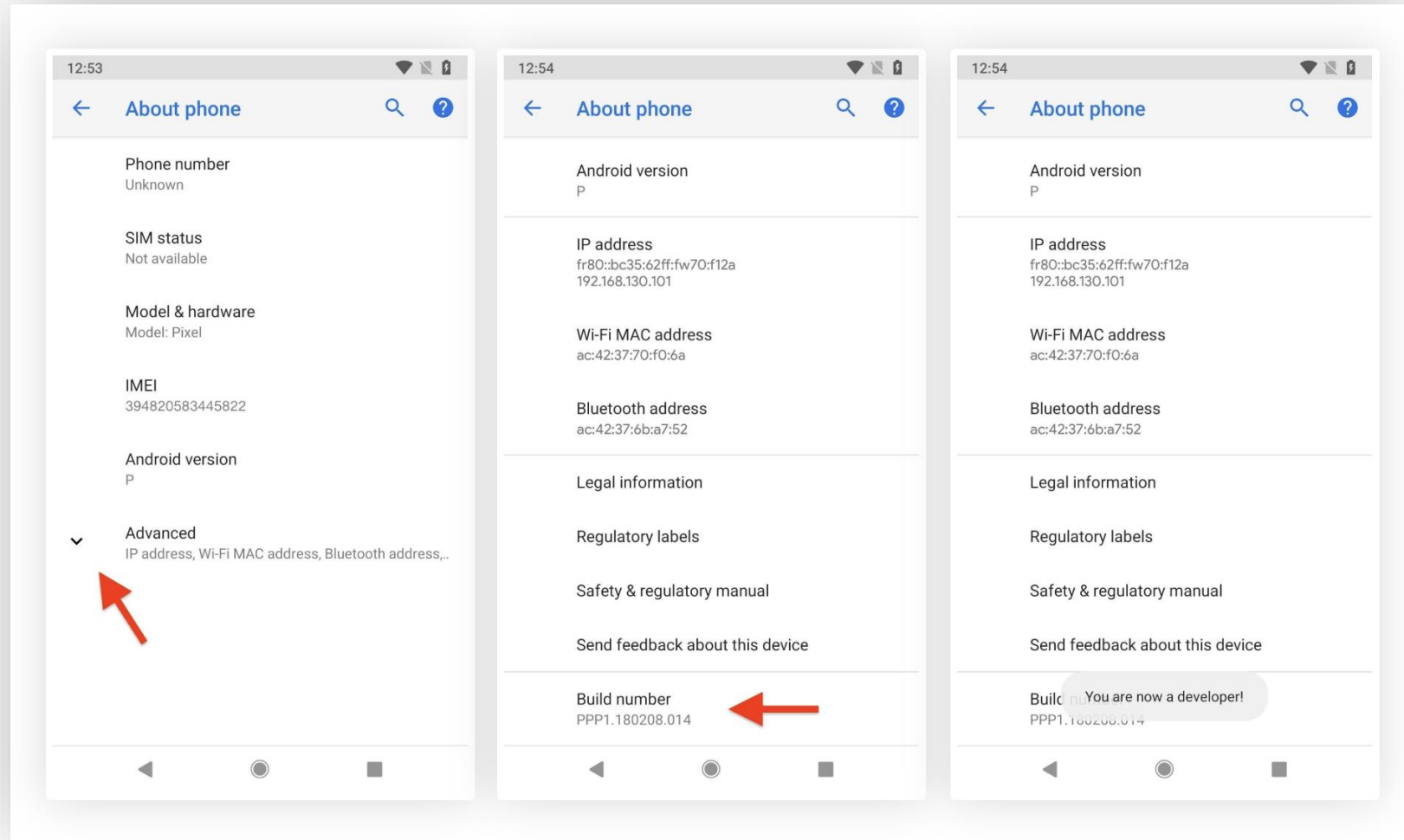


Misc.

Всякое-разное



Developer options



Эмулятор

Эмуляторы существенно расширяют возможности для отладки.

Можно настроить:

- тип и качество сети
- количество ядер CPU, размер ОЗУ и т.д.
- внутреннее/внешнее хранилища
- состояние сенсоров
- наличие/отсутствие аппаратных фишек

...

<https://developer.android.com/studio/run/emulator-extended-controls>



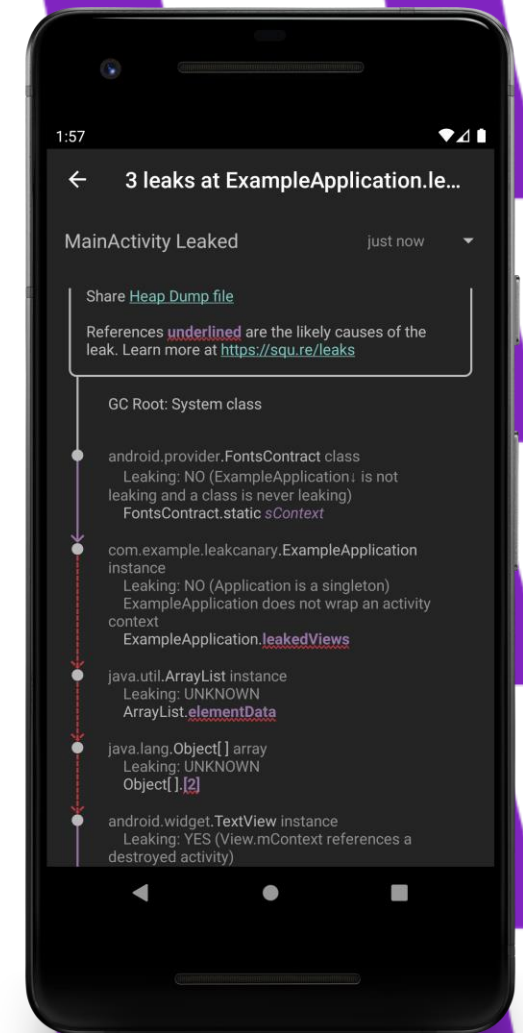
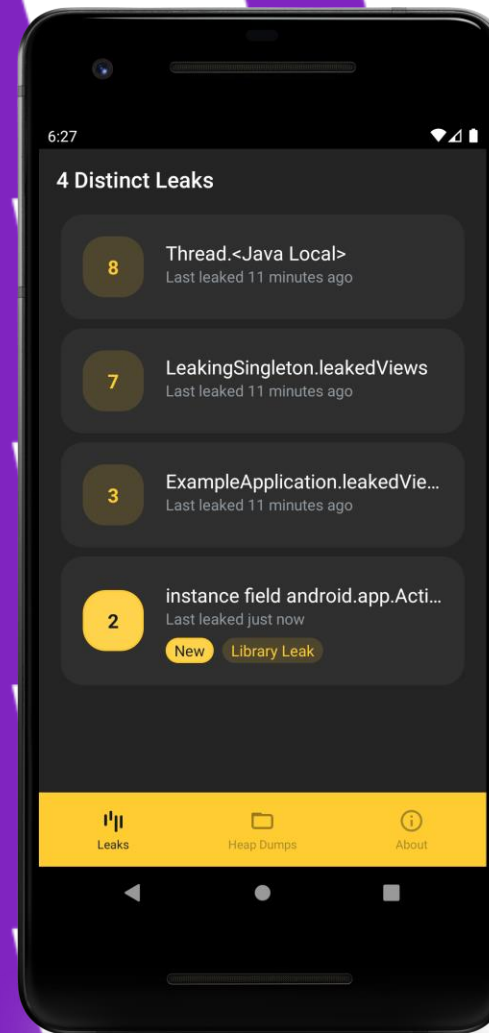


com.squareup.leakcanary:leakcanary-android

Info

LeakCanary automatically detects leaks of the following objects:

- destroyed `Activity` instances
- destroyed `Fragment` instances
- destroyed fragment `View` instances
- cleared `ViewModel` instances
- destroyed `Service` instance

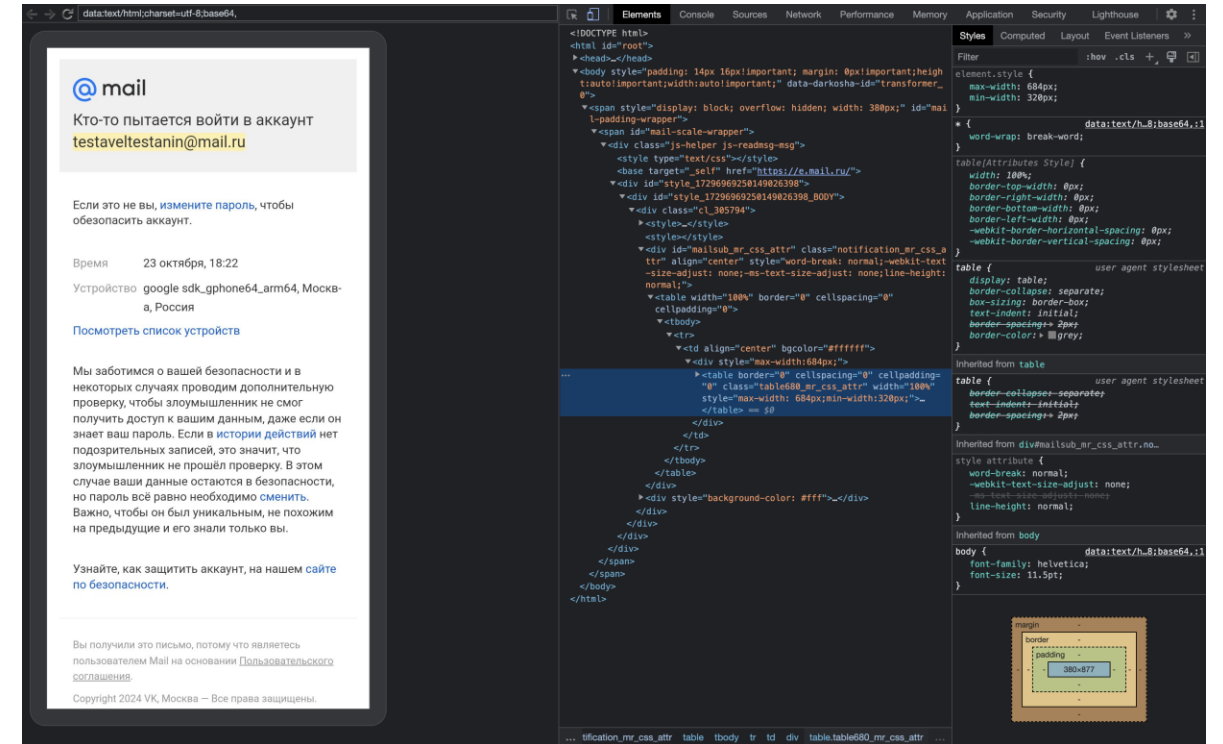
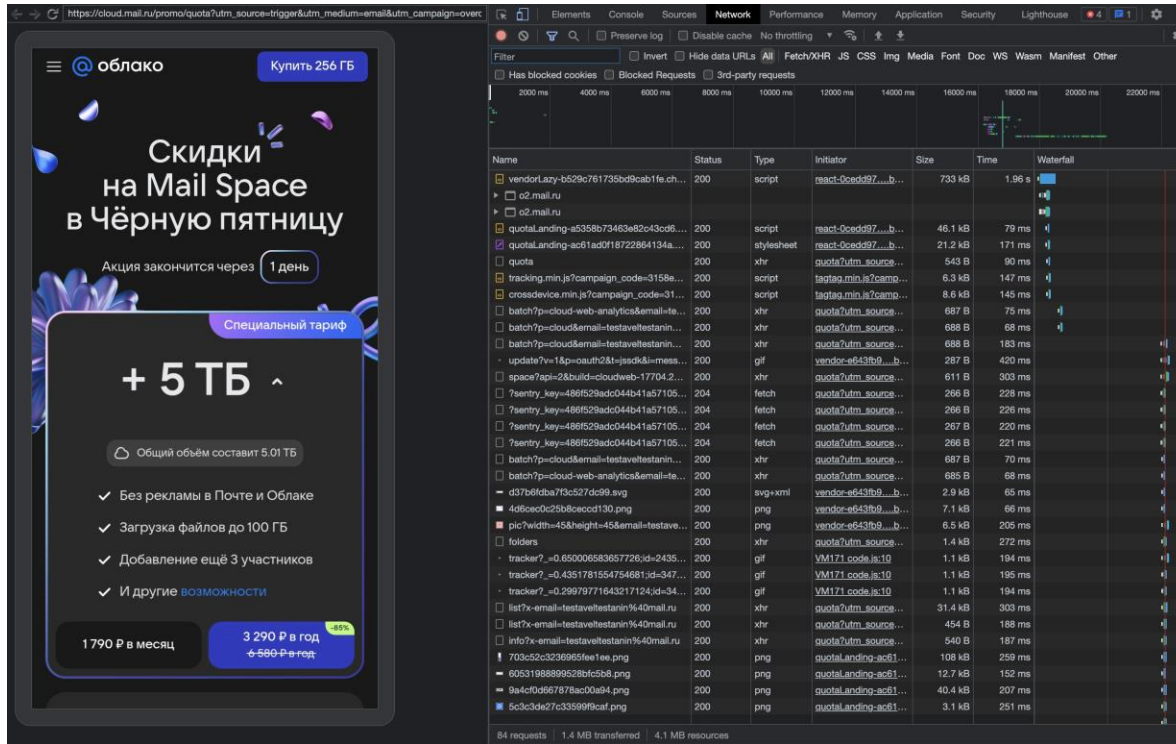


LeakCanary is dumping the memory to investigate leaks.

<https://github.com/square/leakcanary>

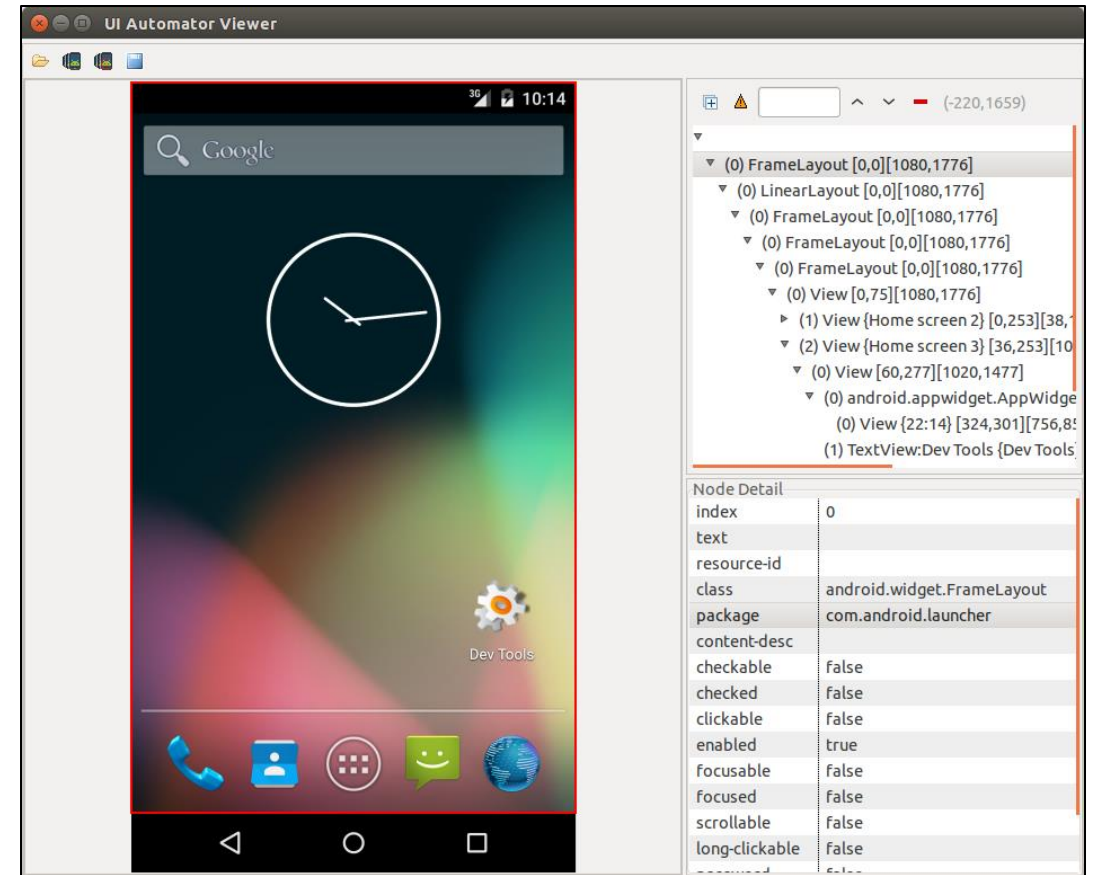
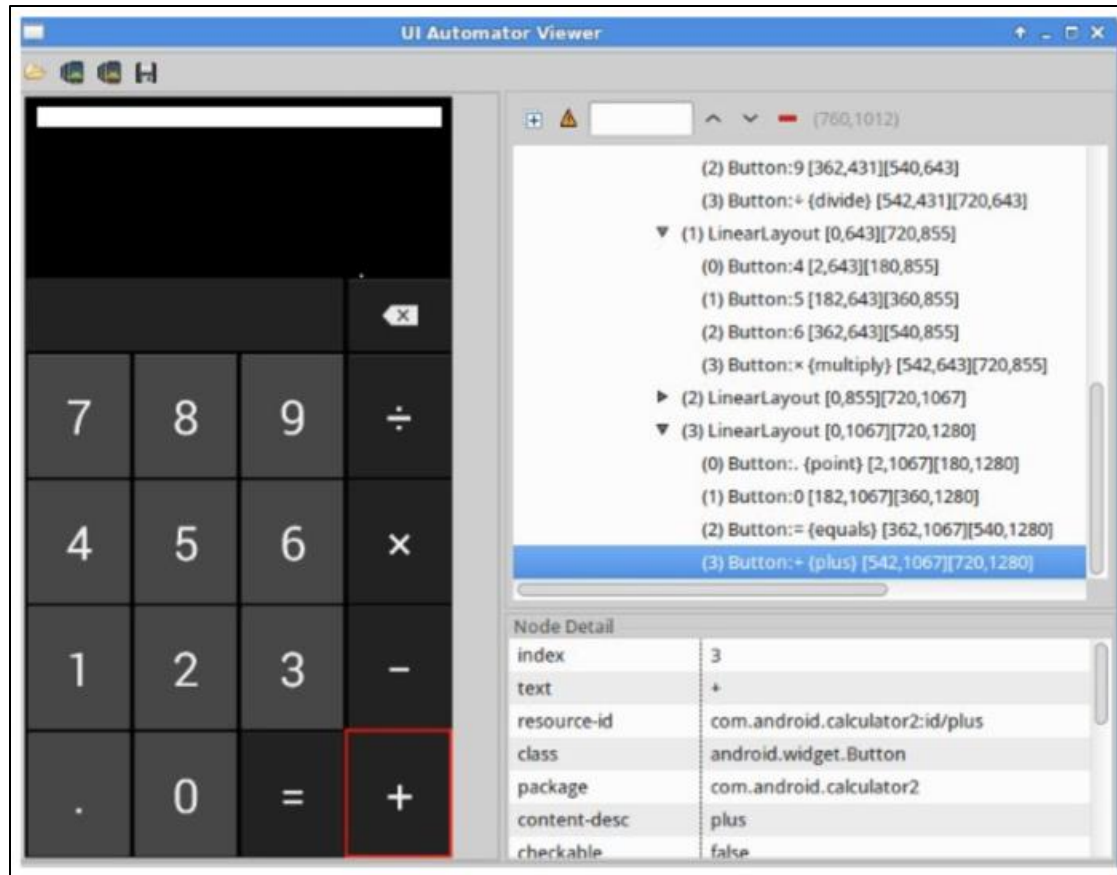
Отладка WebView

chrome://inspect



<https://developer.chrome.com/docs/devtools/remote-debugging/webviews>

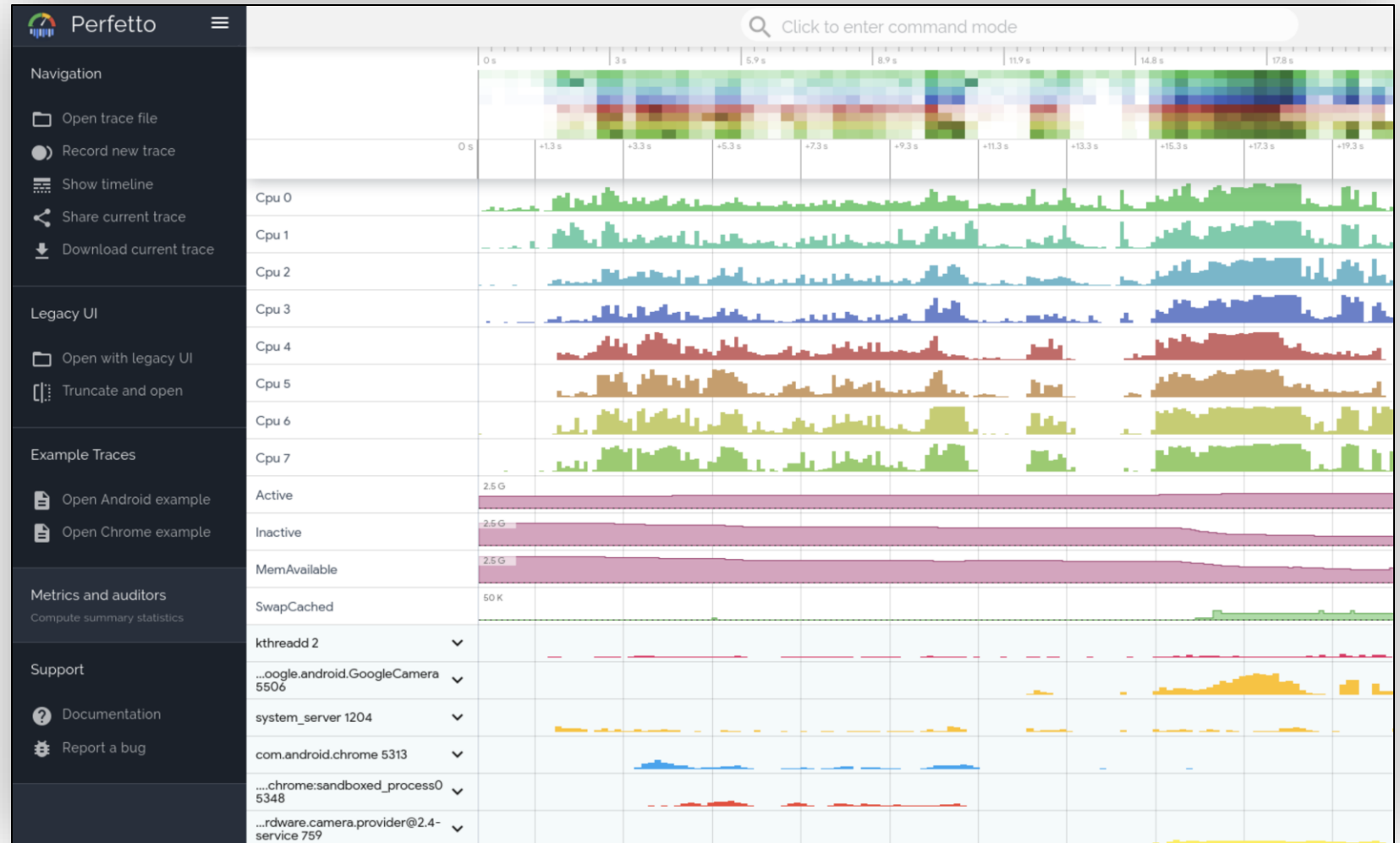
UI Automator Viewer



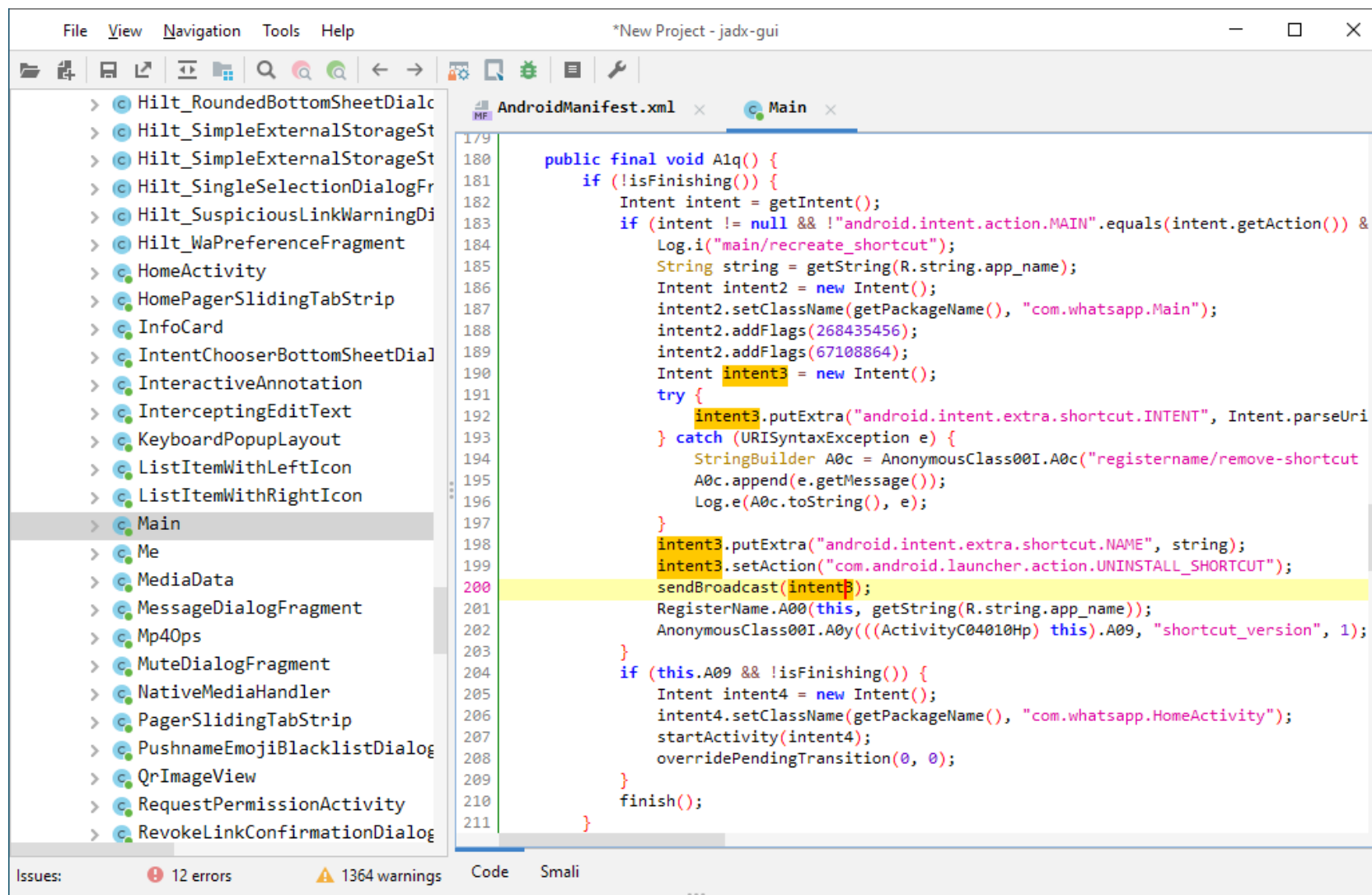
Systrace, Perfetto

Сбор всех событий,
произошедших в системе:

- чем были заняты потоки
- как использовалась память
- операции с диском
- потребление батареи
- ...



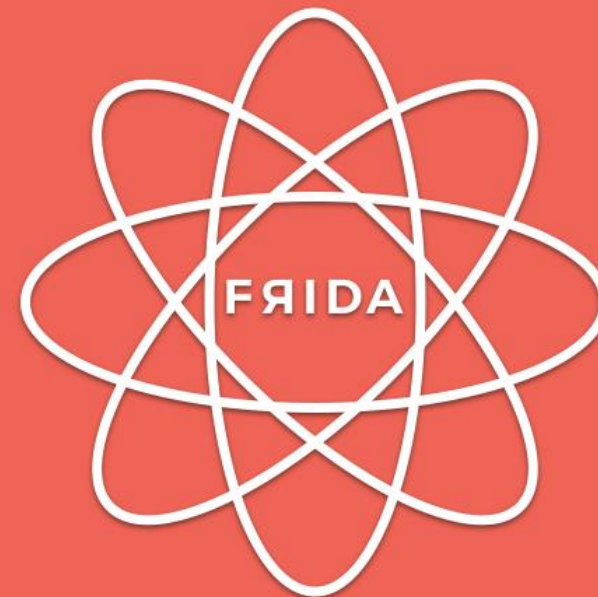
jadx



Frida

Тулкит для [динамической инструментации](#) кода.

«*Greasemonkey for native apps*»



Полезные ссылки

- Как работа с тикетом «Тормозит скролл в RecyclerView» растянулась на три недели ([YouTube](#))
- Что внутри Android SDK (YouTube)
- Рассказ про Frida/Flipper и обзорный доклад ([YouTube](#))
- Дебаг GPU ([YouTube](#))
- Мониторинг приложений в проде ([YouTube](#))
- Как работать с ADB ([YouTube](#))

Спасибо за
внимание!

